

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: B.Tech Branch : Electronics & Telecommunication Engineering and Allied****Semester: VI****Subject Code & Name: BTETPE603A Microprocessors and Microcontrollers****Max Marks: 60****Date: 23/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which pin of the 8051 is used for providing the clock input? a. XTAL1 and XTAL2 b. RST c. PSEN d. ALE	CO1	1
2	The 8051 microcontroller has how many I/O ports? a. 4 b. 2 c. 8 d. 6	CO1	1
3	How much on-chip RAM is available in the standard 8051 microcontroller? a. 64 bytes b. 256 bytes c. 512 bytes d. 128 bytes	CO1	1
4	The instruction MOV A, @R1 uses which addressing mode? a. Immediate b. Direct c. Indirect d. Register	CO1	1
5	Which flag is affected by the CMP instruction in the 8085? a. Auxiliary Carry Flag b. Zero Flag c. Sign Flag d. All of the above	CO1	1
6	What is the function of the HLT instruction in the 8085? a. Halt the program execution permanently b. Stop the microprocessor until reset c. Restart the program execution d. Enable the interrupt system	CO1	1
7	Which of the following instructions is used to copy data from memory to the accumulator in 8085? a. LDA b. STA c. MOV d. MVI	CO1	1
8	Which logical instruction complements all bits of the accumulator in 8085? a. CMA b. CMC c. XRA d. RLC	CO1	1

9	What is the purpose of the CALL instruction?				CO1	1
	a. Halt the program execution	b. Jump unconditionally to a specified address	c. Call a subroutine and save the return address on the stack	d. Skip the next instruction		
10	Which instruction is used to push data onto the stack in 8085?				CO1	1
	a. POP	b. PUSH	c. CALL	d. HLT		
11	Which interrupt in 8085 has the highest priority?				CO1	1
	a. RST 7.5	b. RST 6.5	c. RST 5.5	d. TRAP		
12	In RISC processors, most instructions are				CO1	1
	a. Implemented using microcode	b. Executed in multiple cycles	c. Designed for hardware implementation	d. Supported with variable lengths		
Q. 2	Solve the following.					12
A)	Explain functions of registers in 8085 microprocessor.				CO2	6
B)	Explain the functions of each pin of 8085 microprocessors.				CO3	6
Q.3	Solve the following.					12
A)	Write an 8085-assembly program to perform addition of two 8-bit numbers and store 16-bit result after addition in a specific memory locations.				CO2	6
B)	Write an 8085-assembly program to perform logical AND operation between two data bytes and store the result.				CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Describe the internal memory organization of the 8051 microcontrollers.				CO1	6
B)	What are the key features of the 8051 microcontroller and explain the architecture of the 8051 microcontrollers.				CO1	6
C)	Define Addressing Mode? and explain addressing modes of 8051 microcontroller with examples.				CO3	6
Q.5	Solve Any Two of the following.					12
A)	Write a program to generate a square wave of 50% duty cycle on the pin P1.5 of 8051. Use Timer 0 to generate the time delay.				CO2	6
B)	Describe functions of all bits of TCON register.				CO1	6
C)	Explain the basic concept and working of timers in the 8051 microcontrollers.				CO1	6

Q. 6	Solve Any Two of the following.		12
A)	Describe the functional blocks of the 8255 Programmable Peripheral Interface and explain how to interface it with the 8085 microprocessors.	CO3	6
B)	Draw the interfacing diagram of a 4K EPROM having a starting address 2000H with 8085 microprocessor using 3 to 8 Decoder (74LS138).	CO2	6
C)	Explain the working of an ADC and describe how it is interfaced with the 8051 microcontroller.	CO3	6
*** End ***			